# Nguyen Xuan Hoang

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## Education

**Vo Nguyen Giap High School for Gifted Student** Sep 2022 - May 2025

**VNUHCM – University of Science,** BS in Artificial Intelligence Sep 2025 - Sep 2029

**Summer in Engineering and Applied Sciences (SEAS)** — Quang Tri, Vietnam Jul - Aug 2025

*Participant (selected 43 out of 400 applicants nationwide)* | seas-cvn.com

- Completed an intensive 2-week full-day program on **Artificial Intelligence and Applications**, covering Python programming, linear algebra, and machine learning (adapted from the MIT Computer Science undergraduate curriculum). The program was **modeled after real academic research**, emphasizing problem-driven learning and teamwork.
- Collaborated on a group project applying AI to **solving quantum many-body physics problems**, the initial results show that the model can efficiently find the lowest energy of the system, opening up research directions for various types of quantum systems.
- **Presented project findings** at the conclusion of the program to mentors and peers.
- Worked directly with mentors from **Harvard, MIT, CERN, Stony Brook University, Ericsson Research, UIUC, UC Irvine, VinAI**, and other leading institutions.

## Project

Neural Networks for quantum many-body physics **- Summer in Engineering and Applied Sciences (SEAS)**

- A project combining artificial intelligence and quantum computing to solve one of the hardest problems in physics: finding the lowest energy of a quantum many-body system, especially applied to superconducting materials.
- The research method is based on the Heisenberg model in quantum mechanics to describe the state of many-body systems. The problem is reduced to finding the minimum value of an energy equation through mathematical transformations, with the support of artificial neural networks to predict and approximate the energy matrix.
- In this project, we utilized Google Colab alongside the NetKet library to explore the arrangement of qubits in one, two, and three dimensions. We implemented neural network models and experimented with alternative approaches to address the problem effectively.

## Skills

**Technical:** Python, C++
**Soft skills:** Teamwork, Presentation Skill

## Experience

**Google Cloud AI Study Jam: QuanQuanGCP Season 7** Sep - Oct 2025

*Participant who done 15 badges are listed in study jam (11 skill badges and 4 regular badges)* | profile